

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
M.Tech Degree S2 (S,FE) Examination December 2024 (2022 scheme)

Course Code & Name: 222TCS002 DEEP LEARNING

Max. Marks: 60

Duration: 2.5 Hours

PART A

Answer all questions. Each question carries 5 marks

Marks

- 1 Apply the Adam optimization algorithm to minimize a loss function given an initial weight vector. (5)
- 2 Consider a Convolutional Neural Network (CNN) being used to classify handwritten digits from the MNIST dataset. The network has an input image size of 28x28x1 (grayscale image), a convolutional layer with 32 filters, each 3x3, and a stride of 1. The padding is set to 'same'. (5)

Question:

- a) Calculate the feature map's output size After applying this convolutional layer to the input image.
- b) Explain why padding is used in this context.
- .
- 3 Describe the vanishing gradient problem in Recurrent Neural Networks (RNNs) (5)
- 4 Distinguish between the Skip-gram and Continuous Bag of Words (CBOW) models used in Word2Vec for word embedding. (5)
- 5 Explain self-attention in the transformer architecture. (5)

PART B

Answer any 5 questions. Each question carries 7 marks.

- 6 Analyze the challenges in optimizing neural networks, such as saddle points and exploding gradients (7)
- 7 Construct the AlexNet architecture, discussing its main components and innovations that contributed to its success in deep learning. (7)
- 8 Evaluate the performance of Recurrent Neural Networks (RNNs) for modeling sequences. (7)

- 9 Analyze the use of Bidirectional Encoder Representations from Transformers (BERT) in pretraining for natural language processing tasks, providing a case study on sentiment analysis. (7)
- 10 Explain the positional encoding mechanism in transformer architecture. (7)
- 11 Apply word embeddings using GloVe to your chosen natural language processing task. (7)
- 12 Consider a simple Artificial Neural Network (ANN) with one input, one output, and no hidden layers. The network is trained using the Gradient Descent algorithm. The network's weight is initialized to $w=0.5$, and the learning rate $\alpha=0.1$ (7)

The training data consists of a single data point:

- Input $x=2$
- True output $y=1$

The output of the network is calculated as

$$: \hat{y} = w \cdot x$$

The loss function is the Mean Squared Error (MSE):

$$L = \frac{1}{2} \cdot (\hat{y} - y)^2$$

Question:

1. Calculate the weight update after one iteration of the gradient descent algorithm.
2. Show the complete process, including calculating the gradient and the new weight value.